ECE 491 Image Processing and Computer Vision (Fall 2026)



Lecture 23 Object Classification

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Code & Models: https://github.com/iVMCL/

NC STATE UNIVERSITY Electrical and Computer Engineering

Outline

- **Topics:** benchmarks and backbones; data augmentation; the training recipe (normalization, regularization, optimization); fine-tuning; evaluating a classifier honestly.
- **Key concept:** architecture matters, but so do the data, the recipe and the evaluation protocol.
- **Notes:** notebook 27 (marimo): augmentation explorer, recipe experiment, fine-tuning, confusion matrix, calibration, Grad-CAM.

So far, Image Classification

Classification



CAT

Challenges: Background Clutter



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Image Source: http://cs231n.stanford.edu/slides/2017/cs231n_2017_lecture2.pdf

Challenges: Intra-class Variation



Image Source: http://cs231n.stanford.edu/slides/2017/cs231n_2017_lecture2.pdf

- Another example: think about how many types of chairs in the Hunt library?

Challenges: Scale



slide credit: Fei-Fei, Fergus & Torralba

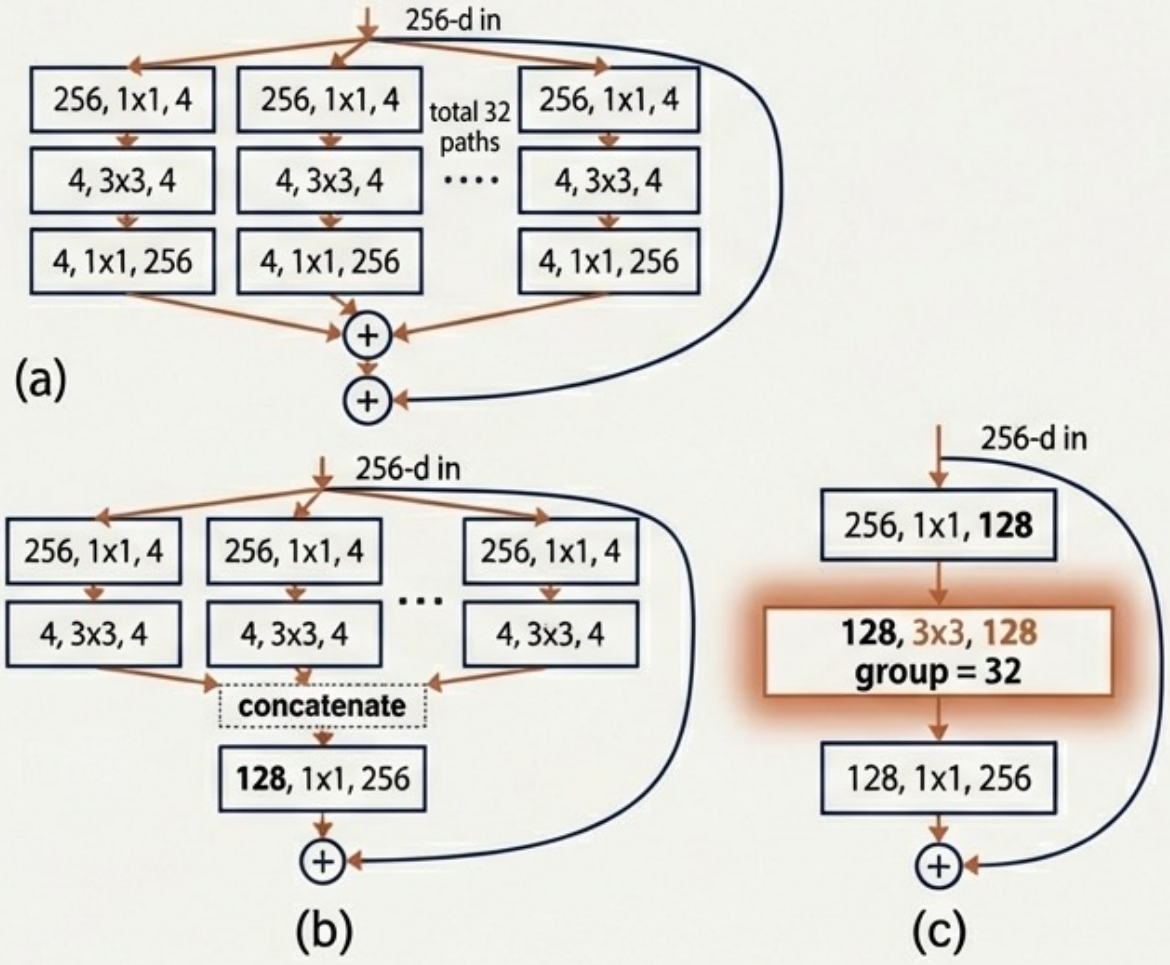
Benchmarks for Image Classification

- **MNIST (1998):** 60,000 training images, 28x28 gray digits, 10 classes.
- **CIFAR-10 / CIFAR-100 (2009):** 50,000 training images, 32x32 color, 10 / 100 classes.
- **ImageNet-1k (ILSVRC, 2012):** 1.28M training photos, 1,000 classes. Winning top-5 error: 25.8% (2011, hand-crafted features), 15.3% (2012, AlexNet), 3.6% (2015, ResNet).
- **A score is a measurement on one test set:** on a newly collected ImageNet test set, every model lost 11–14 points of accuracy; the ranking held (Recht et al., ICML 2019).

Choosing a Backbone (ImageNet top-1)

- **MobileNetV3-L:** 5.5M params, 0.2 GFLOPs, 74.0%
- **ResNet-18:** 11.7M params, 1.8 GFLOPs, 69.8%
- **ResNet-50:** 25.6M params, 4.1 GFLOPs: 76.1% (2016 recipe) vs. 80.9% (2021 recipe), same network
- **ConvNeXt-T:** 28.6M params, 4.5 GFLOPs, 82.5%
- **ViT-B/16:** 86.6M params, 17.6 GFLOPs, 81.1%
- The training recipe alone moves ResNet-50 by 4.7 points, more than separates it from any newer architecture in this list.

Refining the Heuristic with Group Convolutions



ResNeXt (2016)

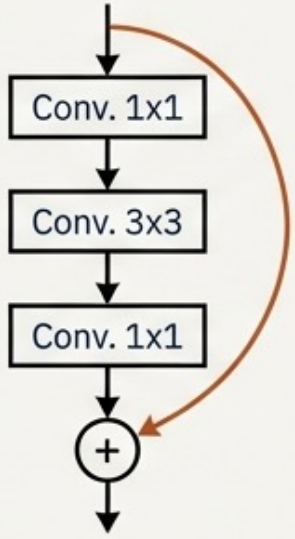
The Limitation of Inception:
Modules were heavily hand-crafted stage-by-stage with customized filter sizes.

The Breakthrough:
Stacking **Split-Transform-Aggregate** blocks of the exact same shape using **Group Convolutions**. It **standardizes the Inception heuristic** inside a ResNet shell for automated scalability.

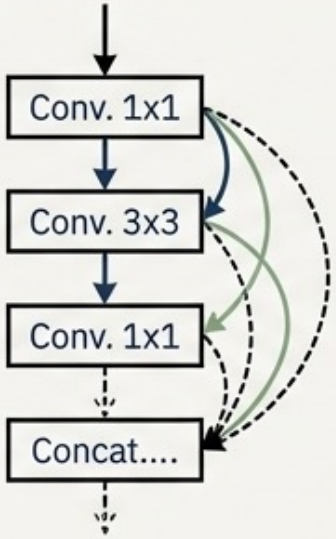
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Balancing Feature Exploration and Exploitation

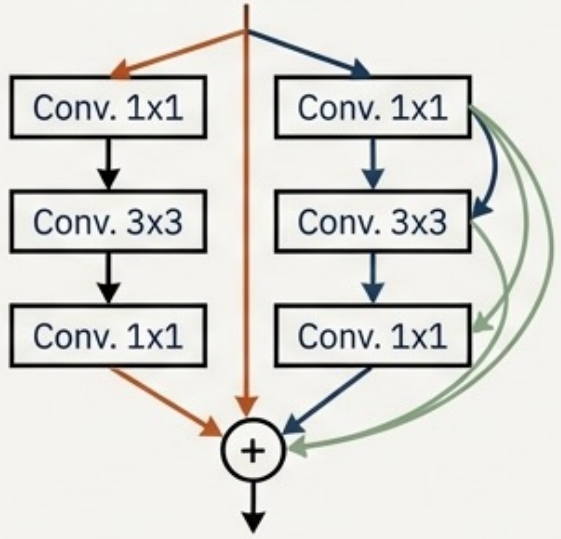
Residual Network



Densely Connected Network



Dual Path Architecture



ResNet

Focuses on feature exploration via residual paths.

DenseNet

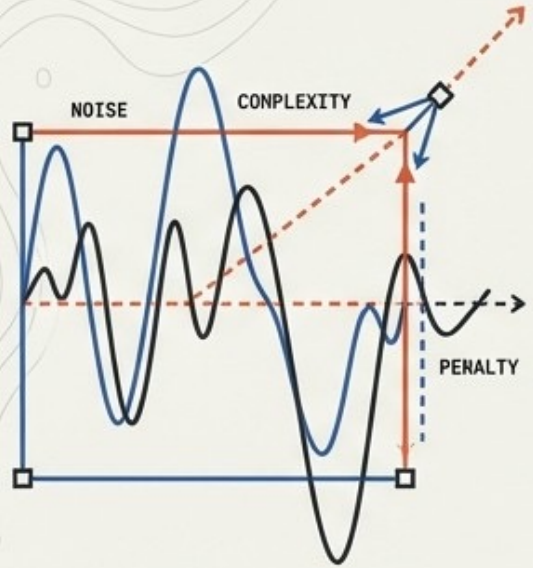
Focuses heavily on feature reuse by densely concatenating connections.

Dual Path Networks (DPN, 2017)

Wires ResNet and DenseNet paths in parallel, creating a macro-architecture that achieves a perfect balance of exploration and exploitation.

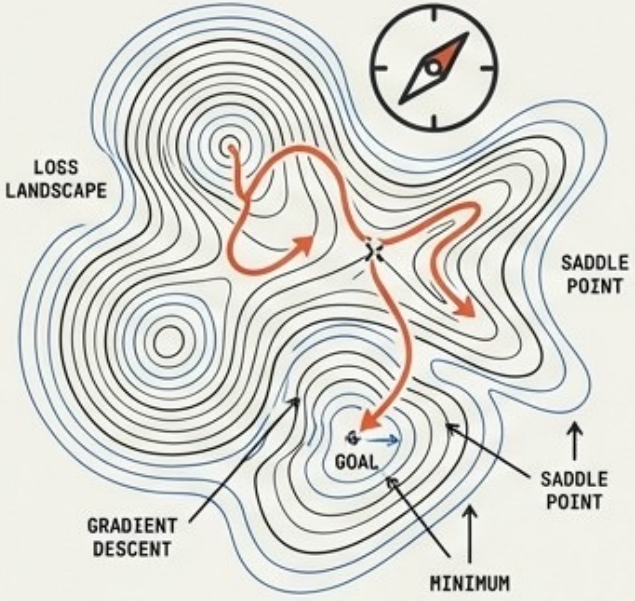
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The Three Pillars of Training



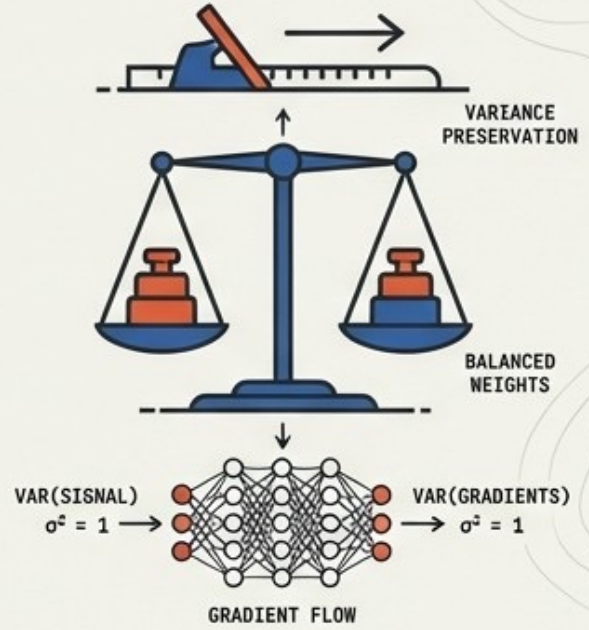
1. REGULARIZATION

The Constraints. Preventing the model from memorizing noise by penalizing complexity.



2. OPTIMIZATION

The Navigation. Guiding the weights through the saddle points and ravines of the loss landscape.



3. INITIALIZATION

The Start. Preserving signal variance to ensure gradients survive the deep network.

Goal: Train a generalized, robust Deep Neural Network.

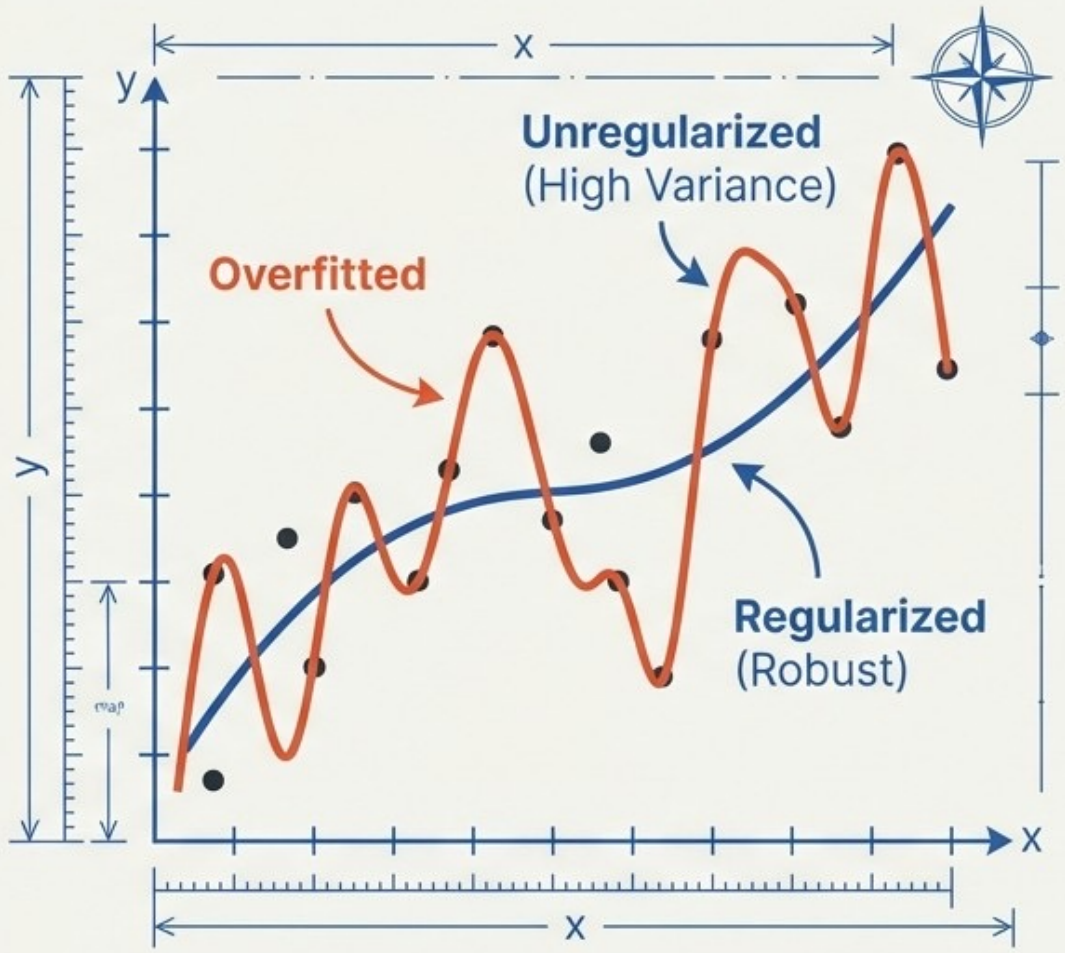
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Regularization: The Problem of Overfitting

Deep Neural Networks are powerful function approximators. Left unchecked, they do not learn patterns; they memorize noise.

$$\text{Loss}(\theta) = \frac{1}{N}\sum L(f(x), y) + \lambda \cdot R(\theta)$$

Penalty for Complexity



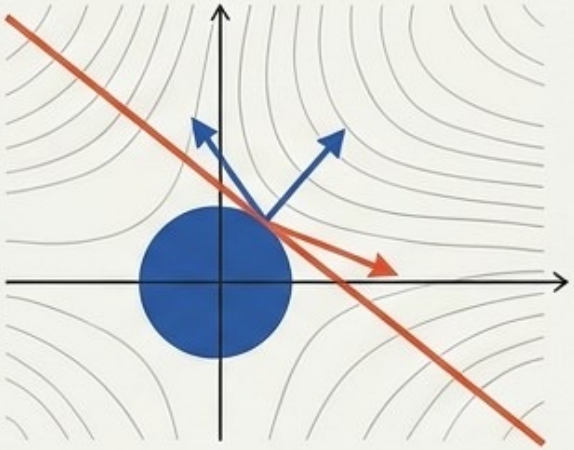
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Penalizing Complexity

L2 Regularization (Weight Decay)

$$R(W) = \sum W^2$$

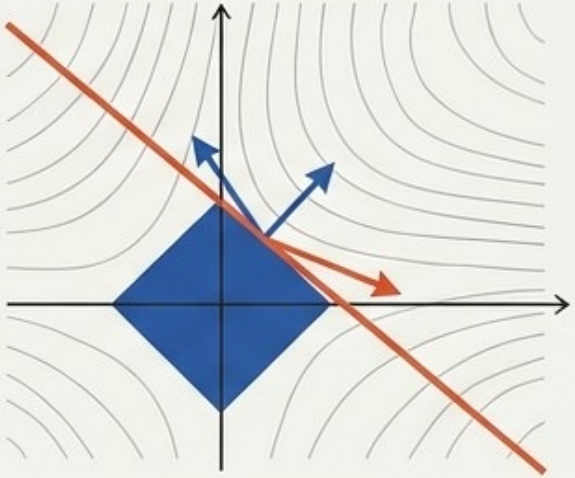
Penalizes the square of weights. Discourages large peaks, spreading energy diffusely across the matrix.



L1 Regularization (Lasso)

$$R(W) = \sum |W|$$

Penalizes absolute value. Drives weights to exactly zero, creating sparsity (feature selection).

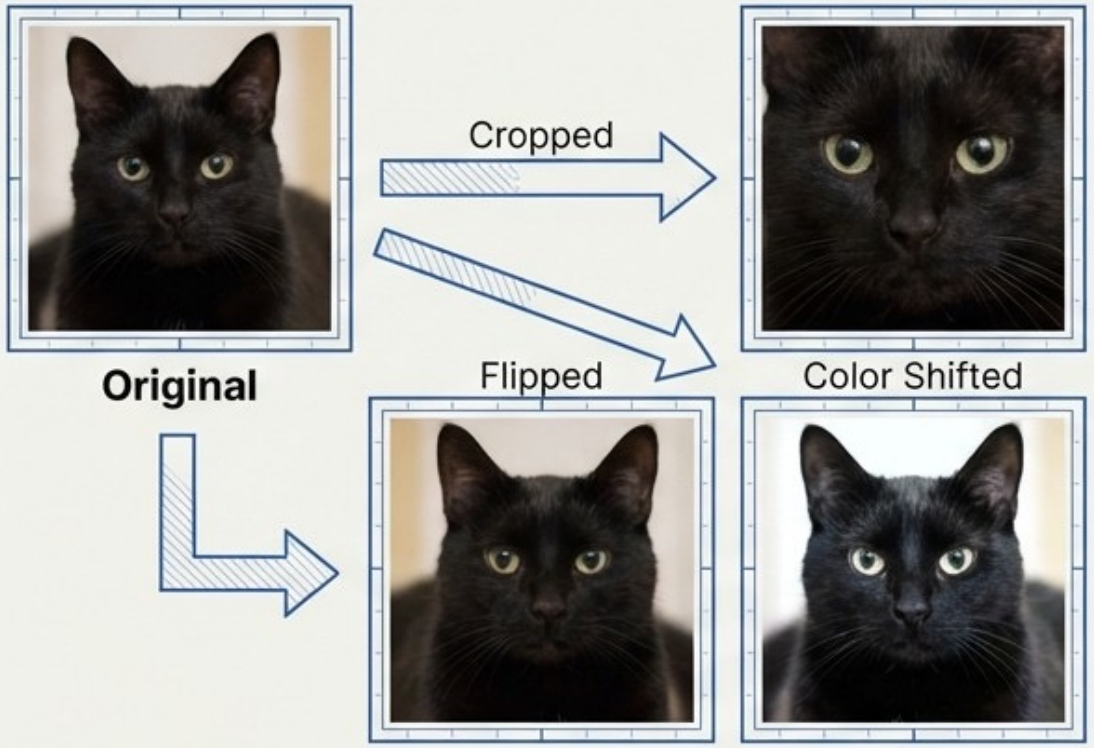


Elastic Net combines both: $\lambda_1|W| + \lambda_2 W^2$

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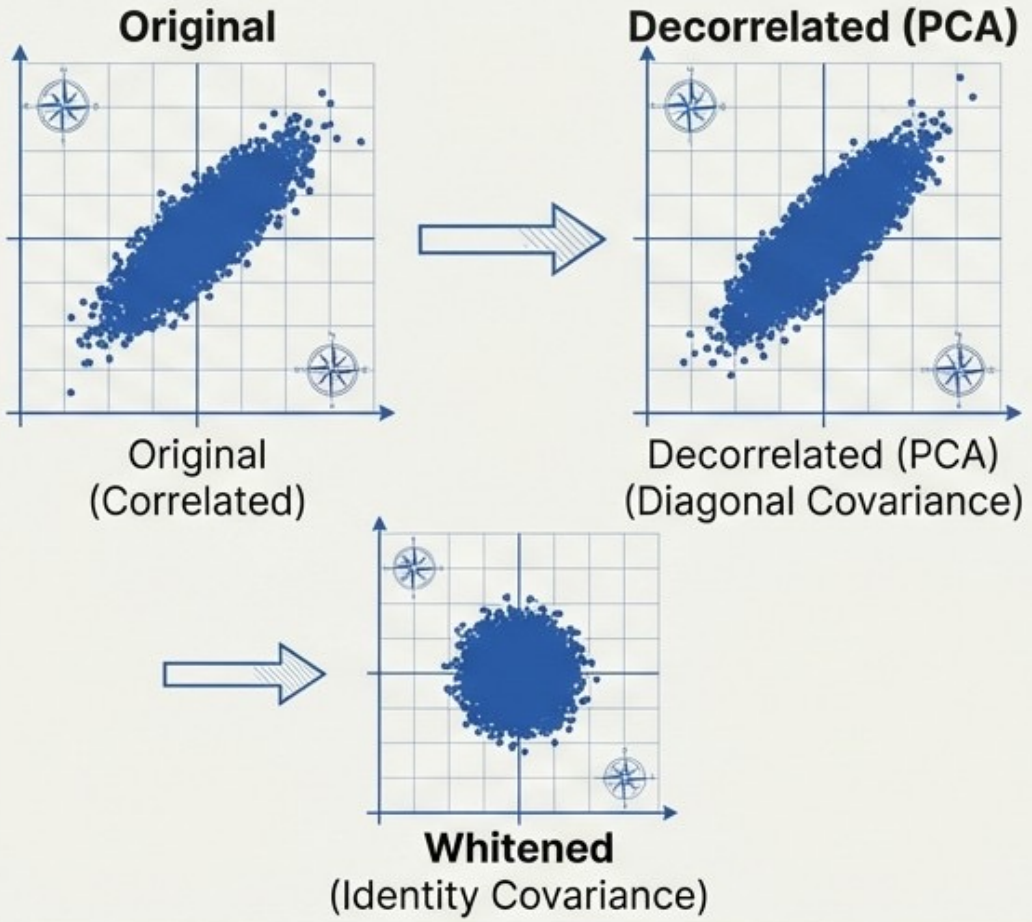
Data-Side Regularization

Augmentation



Forcing invariance to translation, rotation, and lighting.

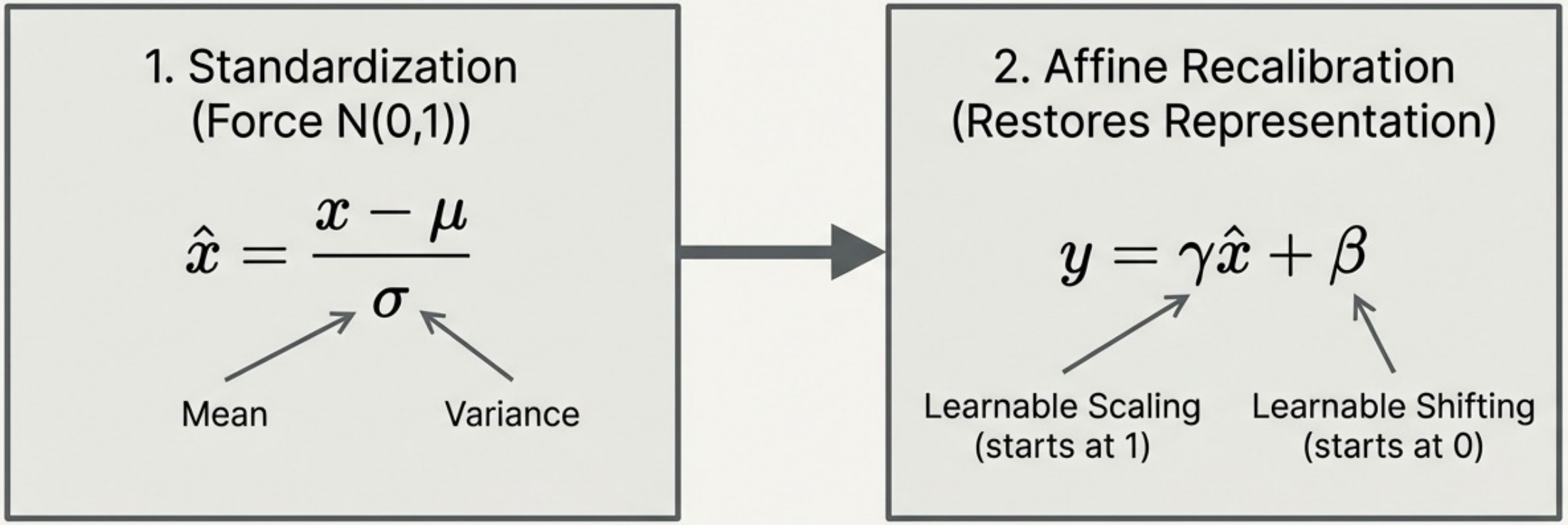
Preprocessing (Whitening)



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The Mechanism

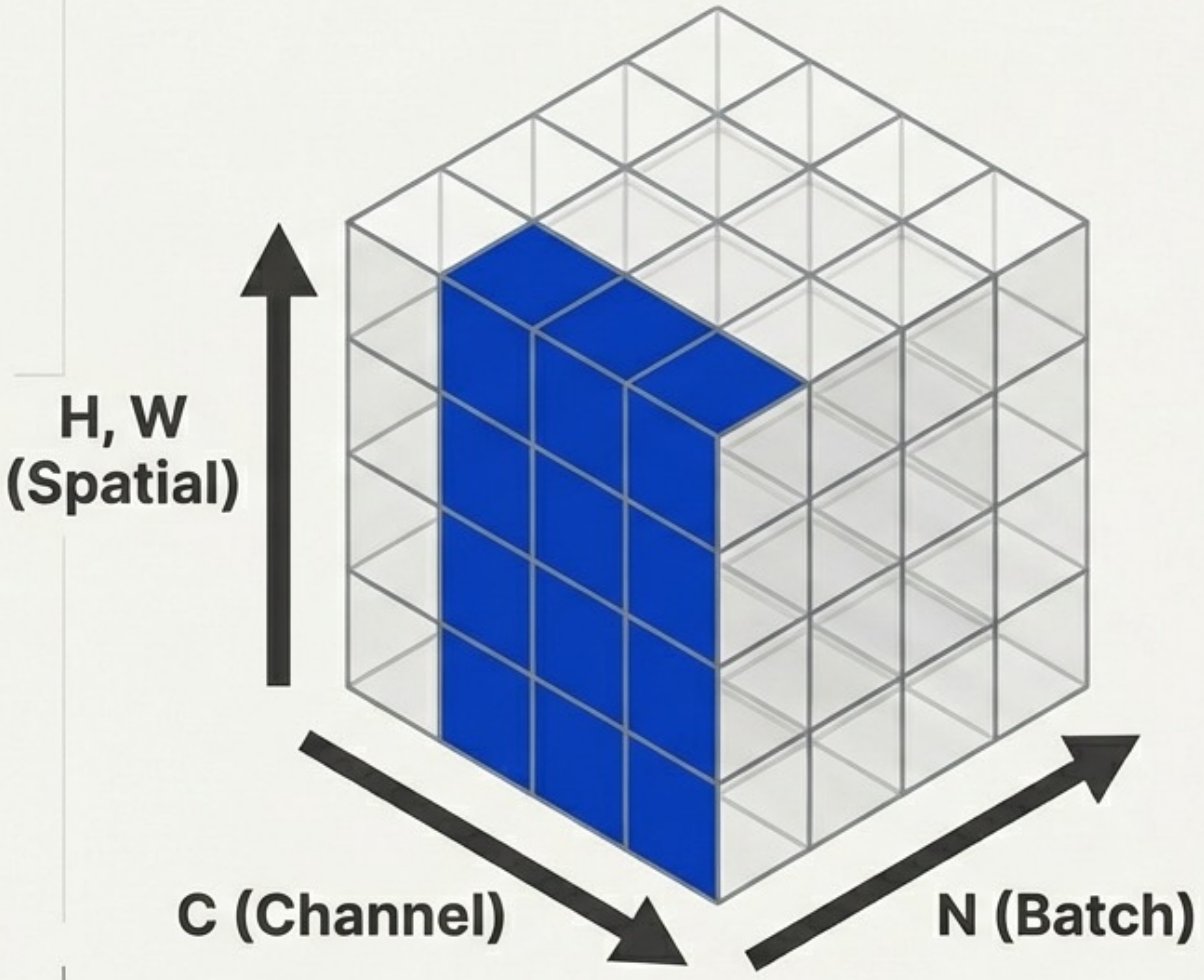
Standardization + Recalibration



Crucial Insight: The network can ‘undo’ the normalization if the task requires it.

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Batch Normalization (BN)



- **Dimensions**: (N, H, W)
- **Status**: The Industry Standard for Classification.
- **Pros**: Stabilizes training, allows higher learning rates.
- **Cons**: Dependent on Batch Size. Inconsistent Training vs. Inference behavior.
- **Source**: Ioffe & Szegedy, 2015.

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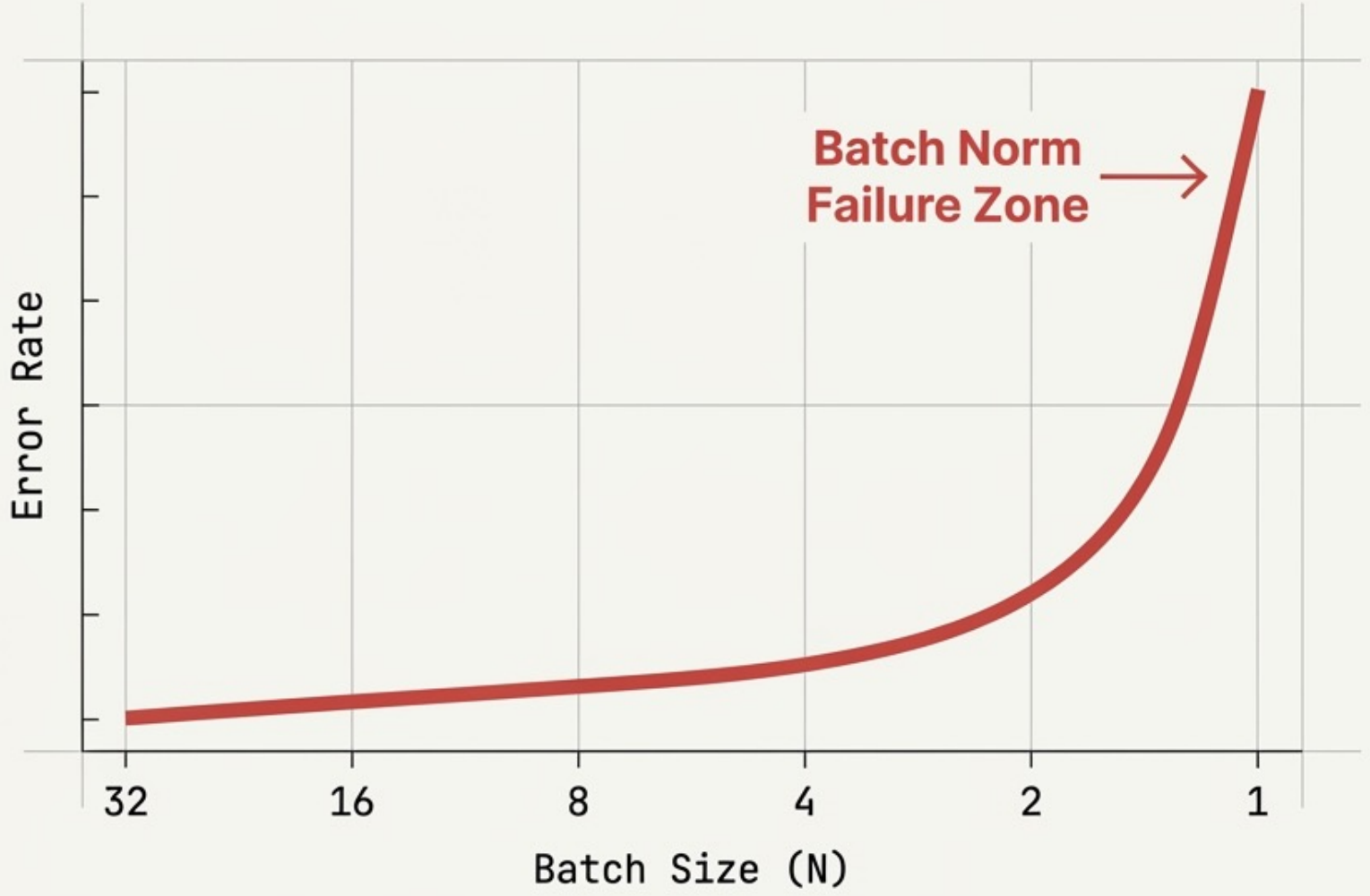
The Small Batch Dilemma

Problem: Tasks like Object Detection and Segmentation require high-resolution images, consuming massive GPU memory.

Constraint: This forces Batch Size to be small (N=1 or 2).

Result: Batch statistics (μ, σ) become noisy estimates, destabilizing training.

Reference text:
“The real question is how to deal with Batch_size = 1”



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Selection Guide: Choosing the Right Norm

Batch Norm (BN)



- **Use Case:** Classification (Standard)
- **Condition:** Batch Size ≥ 32

Group Norm (GN)



- **Use Case:** Detection / Segmentation
- **Condition:** Batch Size < 8

Layer Norm (LN)



- **Use Case:** Sequential / NLP
- **Condition:** Transformers / RNNs

Instance Norm (IN)



- **Use Case:** Generative / Style
- **Condition:** GANs / Style Transfer

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Solution 2: Adam

Adaptive Moment Estimation



- **Adaptive Learning Rates**: Scales the step size **inversely** to the variance. Large gradients? Take smaller steps. Flat terrain? Take larger steps.
- **Bias Correction**: Essential for the first few steps to prevent estimates from starting at zero.

The Practitioner's Default: Adam (LR = 3e-4)

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The Practitioner's Default Config

A safe starting point for Deep Learning

```
config = {
    "ARCHITECTURE": {
        "Activation": "ReLU or GELU",
        "Initialization": "He Initialization (k=2/n)"
    },
    "OPTIMIZATION": {
        "Optimizer": "Adam",
        "Learning Rate": 3e-4,  # Safe start
        "Scheduler": "Cosine Decay"
    },
    "REGULARIZATION": {
        "Weights": "Weight Decay (L2)",
        "Data": "Augmentation (RandomCrop, Flip)"
}
```

Start here. Then tune based on the terrain.

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Notebook 27: What the Recipe Buys

- Same 36k-parameter ConvNet, 2,000 CIFAR-10 training images, 30 epochs, same optimizer and seed.
- **Plain:** train 55%, test 47%. It cannot even fit its training set.
- **+ BatchNorm:** train 96%, test 59%. It trains, but memorizes: a 35-point train–validation gap.
- **+ BatchNorm + augmentation (flip, crop):** train 73%, test 61%. The gap shrinks to 11 points.
- Normalization helps optimization; augmentation helps generalization.

Fine-Tuning a Pretrained Network

- **Linear probe:** freeze the pretrained backbone; train only a new last layer.
- **Fine-tuning:** replace the last layer and keep training all layers, with a backbone learning rate about 10x smaller than the new layer's.
- **Notebook 27 (ResNet-18, 2,000 CIFAR-10 images, 64x64):** from scratch 61%, linear probe 67%, fine-tuning 73% (backbone lr 1e-4).
- **Too large a learning rate destroys the pretrained features:** backbone lr 1e-3 gives 69%; lr 1e-2 (a from-scratch rate) only 19%, below training from scratch.

Evaluating a Classifier Honestly

- **Three disjoint splits:** train (fit the weights), validation (choose everything else), test (report once).
- **Beyond top-1:** top-k, per-class accuracy, the confusion matrix, and the actual mistakes.
- **Notebook 27:** top-1 73.4%, top-2 88.4%; per-class 59% (bird) to 86% (frog); most frequent confusions dog→cat (144) and cat→dog (98).
- **Imbalanced classes:** a classifier can be 99% accurate by never predicting a class that is 1% of the data. Report per-class recall.

Calibration: Can We Trust the Confidence?

- **Calibrated:** among predictions made with 80% confidence, about 80% are correct.
- **Reliability diagram and ECE:** accuracy vs. confidence in bins; ECE is the average gap, weighted by bin size.
- **Temperature scaling (Guo et al., ICML 2017):** divide the logits by one T fitted on the validation set. The argmax, and so the accuracy, is unchanged.
- **Notebook 27:** T = 1.15; mean confidence 77.8% → 74.2% (accuracy 73.4%); ECE 4.4% → 1.7%.

Summary and Next Lecture

- Benchmarks (MNIST, CIFAR, ImageNet) make progress measurable; a score is a measurement on one test set.
- The recipe matters as much as the architecture: the same ResNet-50 gains 4.7 ImageNet points from a better recipe.
- Start from a pretrained network: a linear probe as the baseline, then fine-tune with a small learning rate.
- Evaluate honestly: choose on validation, report test once, read the confusion matrix, check calibration and Grad-CAM.
- **Next (L24, Wed 11/11):** object detection, from R-CNN to YOLO.